

Problem Description – Retail Store Inventory Analysis

Retail stores manage a large number of products across different categories, regions, and seasons. Maintaining the right inventory level is important to avoid overstocking, stock shortages, revenue loss, and reduced customer satisfaction. Manual inventory tracking becomes difficult as product volume and sales transactions increase.

This project focuses on analyzing the **Retail Store Inventory dataset** to understand product performance, inventory levels, pricing patterns, customer demand, and sales trends. Using **Pandas, SQL, and data analysis techniques**, the dataset is cleaned, processed, and analyzed to generate meaningful insights.

The analysis helps answer business questions such as:

- Which product categories generate the highest sales?
- Which regions perform better in terms of revenue?
- Are inventory levels sufficient to meet demand?
- How do pricing and competitor pricing affect sales?
- What seasonal factors influence product demand?
- Which products require inventory replenishment?

The main objective of this project is to improve **inventory management, sales performance, demand forecasting, and business decision-making** through data-driven insights.

Tools Used:

- Python (Pandas)
- PostgreSQL / SQL
- Jupyter Notebook
- Power BI (optional for dashboard visualization)